

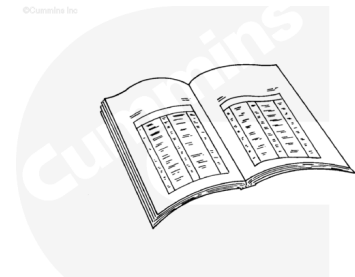
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

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Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries or air supply to the air starter to prevent accidental engine starting. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/89/89-016-005.html>)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-024.html>)

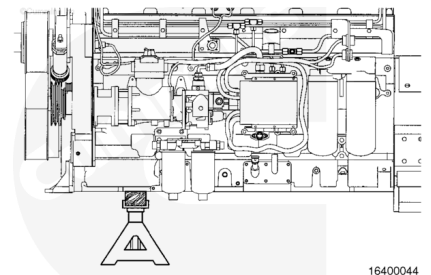
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-020.html>)
- Drain the lubricating oil and remove the oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)
- Remove the stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)

Remove

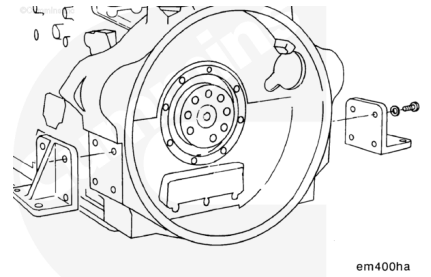
Put a wooden block the width of the engine block skirt between the floor jack and the engine block to prevent damage to the sealing surface.

Use a floor jack or a suitable lifting fixture to support the front of the engine.

Lift and support the rear of the engine with a suitable hoist.

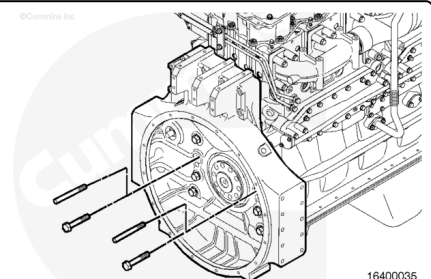


Remove the rear engine mounts from the flywheel housing.



⚠ WARNING ⚠

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Use two guide studs to prevent the flywheel housing from rotating during disassembly.

Remove two capscrews.

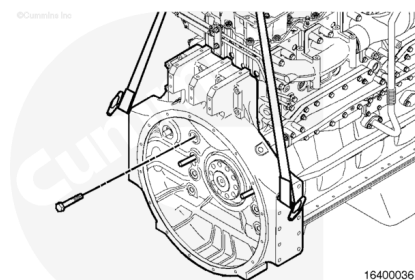
Install the guide studs.

Use a hoist, T-handle, and a lifting sling.

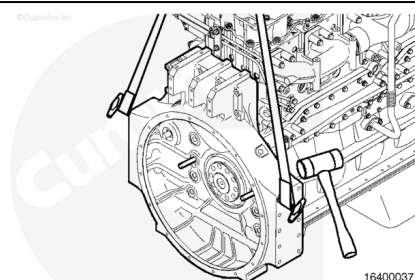
Install the T-handles.

Adjust the hoist until there is tension in the lifting sling.

Remove the remaining capscrews.



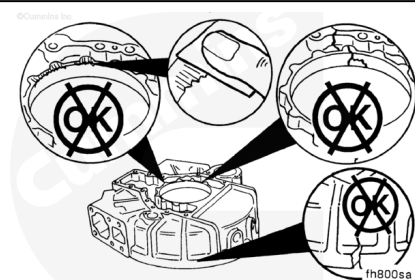
Use a mallet to tap the flywheel housing off the two locating dowels. Remove and discard the gasket material.



Inspect for Reuse

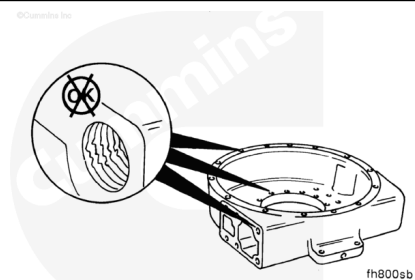
Inspect all surfaces for nicks, burrs, and cracks.

Use a fine crocus cloth to remove small nicks and burrs.



Inspect all threaded capscrew holes for damage.

Repair or replace the housing if the capscrew holes are damaged.

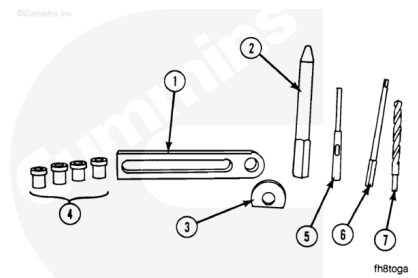


Redowel

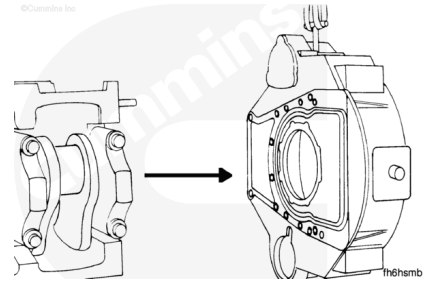
The tools needed to perform this procedure are:

Drill ream fixture, Part No. ST-1232, that contains:

1. Plate, Part No. ST-1232-1
2. Locator pin, Part No. 3375052
3. Spacer washer, Part No. ST-1232-2
4. Drill/ream actual sizes bushing set depends on the dowel size
5. Drill adapter locally obtained; use to adapt open-shank reamers to drill-chuck
6. Reamer locally obtained
7. Drill bit locally obtained.

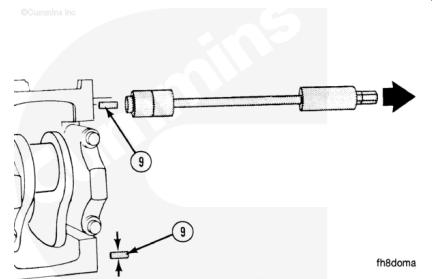


Remove the flywheel housing.



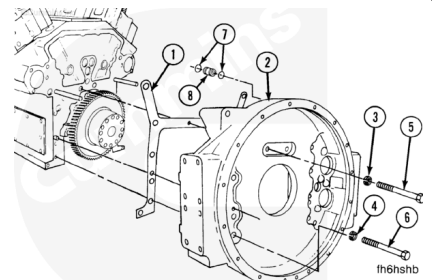
Use a dowel pin extractor, Part No. ST-1134, or equivalent.
Remove the two dowels (9) from the block.

Measure a dowel pin that has been removed so that an
oversize dowel pin can be determined.

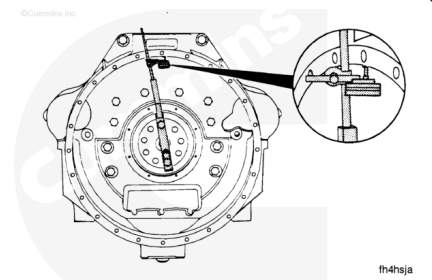


Do **not** tighten the capscrews. The flywheel housing will have
to be aligned.

Install the flywheel housing.



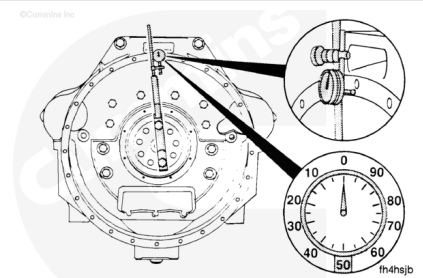
Move the housing with a mallet until the bore is within
specifications. Make sure that the face of the housing is in
alignment.



Measurements				
		mm	in	
Flywheel Housing Inner Diameter	No. 0	648	25.5	
	No. 1	511.2	20.13	
Flywheel Housing - Radial Runout				
		mm	in	
No. 0		0.25	MAX	0.010
No. 1		0.20	MAX	0.008

Measure the flywheel housing face runout.

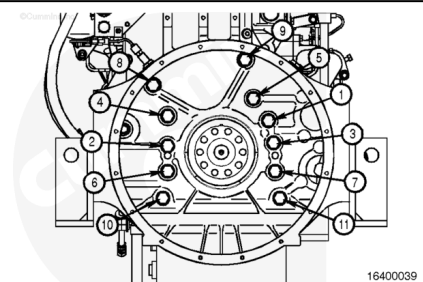
Flywheel Housing - Face Runout				
	mm		in	
No. 0		0.25	MAX	0.010
No. 1		0.20	MAX	0.008



When the housing is in alignment, tighten the capscrews in the sequence shown.

Torque Value:

1. 380 n•m [280 ft-lb]
2. Loosen
3. 410 n•m [302 ft-lb]

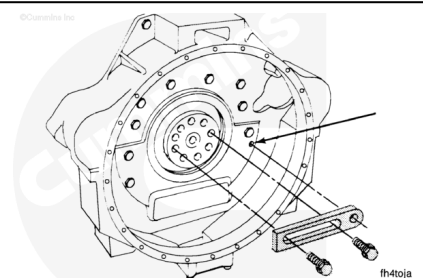


After the capscrews are tightened, check the alignment again.

Use the appropriate capscrews.

Attach the plate, Part No. ST-1232-1, that is contained in the drill ream fixture, Part No. ST-1232, to the crankshaft.

Do **not** tighten the capscrews so much that the plate does **not** move.



Use the locator pin to align the plate with the hole for the dowel pin. Tighten the capscrews. The taper on the pin **must** engage the dowel pin hole.

The locator pin **must** rotate easily after the capscrews are tightened.

The crankshaft **must** be locked in position. It **must not** turn during reaming.

"Lock" the crankshaft in position. Be sure the locator pin is still in alignment and that the locator pin can be rotated easily.

fh4t0ua

Measure the dowel pins to be installed.

Obtain a reamer (5) that is 0.013- to 0.025-mm [0.0005- to 0.001-in] smaller than the dowel.

The dowel **must** protrude from the block by one-half of the flywheel housing wall thickness.

There are three oversize dowel pins available from Cummins Inc.

fh8t0ua

Measurements			
		mm	in
Oversize Dowel Pin - Outer Diameter	0.015-in Oversize	13.08	0.515
	0.030-in Oversize	13.46	0.530
	0.045-in Oversize	13.84	0.545

Install the appropriate drill bushings (4). The following bushings are available from Cummins Inc.

fh4t0jb

Drill/Ream Bushing Sets - 25.4 mm [1 in]	
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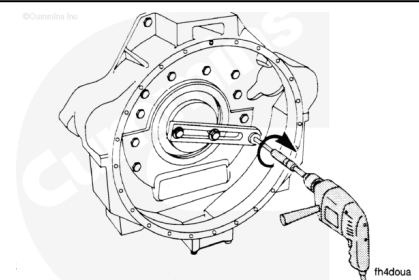
Oversize mm	Oversize [in]	Bushing Size	Bushing Size [mm]	Bushing Size[in]
3376495		Special	12.304	[0.4844]
		Standard	12.700	[0.5000]
0.38	[0.015]		13.096	[0.5156]
0.76	[0.030]		13.492	[0.5312]
1.14	[0.045]		13.879	[0.5464]
ST-1234		Standard	14.288	[0.5625]
0.38	[0.015]		14.684	[0.5781]
0.76	[0.030]		15.080	[0.5937]
1.14	[0.045]		15.479	[0.6094]
ST-1235		Standard	15.875	[0.6250]
0.38	[0.015]		16.271	[0.6406]
0.76	[0.030]		16.667	[0.6562]
1.14	[0.045]		17.066	[0.6719]
ST-1236		Standard	17.463	[0.6875]
0.38	[0.015]		17.859	[0.7031]
0.76	[0.030]		18.255	[0.7187]
1.14	[0.045]		18.654	[0.7344]
ST-1237		Standard	19.050	[0.7500]
0.38	[0.015]		19.446	[0.7656]
0.76	[0.030]		19.842	[0.7812]
ST-1238			22.621	[0.8906]
			23.813	[0.9375]

The drill bushing that is used **must** be the same size as the reamer (or the drill) that is used.

When the new dowel pins are more than 0.38-mm [0.015-in] **larger** than the old dowels, drill the hole to a size that is slightly smaller than the reamer. Then the reamer will **not** have to remove an excessive amount of material.

⚠ CAUTION ⚠

Do not allow metal chips to enter the engine. Damage to the engine will result.



Ream the hole until the reamer touches the bottom of the hole in the block.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

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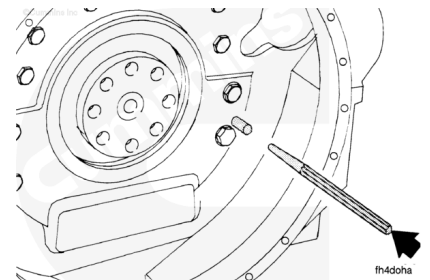
Remove the reamer. Clean the hole with solvent. Dry with compressed air. Push the reamer through the hole again. The reamer **must** touch the bottom of the hole in the block.

After reaming one hole, turn the plate and align it with the next dowel hole. Repeat the procedure in the next hole.

Remove the plate from the crankshaft.

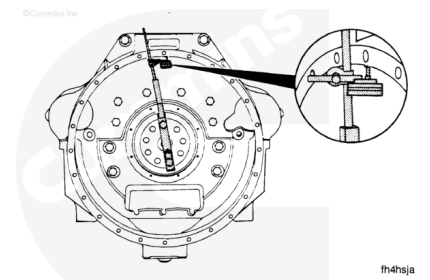
⚠ CAUTION ⚠

The dowel hole must not contain any metal chips. Damage will result.



Use a square-nose drift. Push in each dowel pin until it touches the bottom of the hole in the block.

After the dowels are installed, measure the bore and the face alignment again. The specifications were given earlier in this procedure.

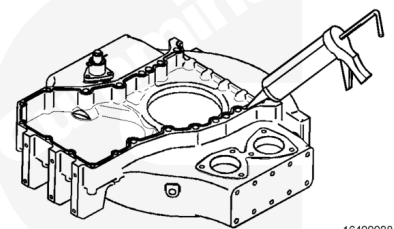


Install

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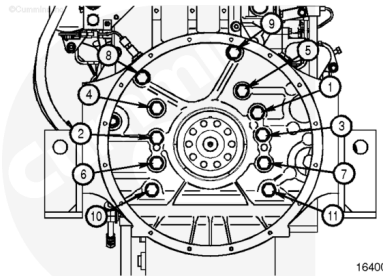


Install liquid gasket material, as shown.

Use guide bolts to help during alignment.

Install the housing and the capscrews.

Tighten the numbered flywheel housing capscrews using the sequence shown.



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Torque Value:

1. 380 n•m [280 ft-lb]
2. Loosen
3. 410 n•m [302 ft-lb]

Tighten the remaining flywheel housing capscrews **not** mentioned in sequence.

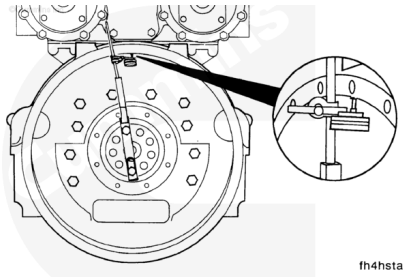
Torque Value: 100 n•m [74 ft-lb]

Measure the flywheel housing alignment.

The bore and the face of the housing **must** align with the crankshaft.

The indicator arm **must** be rigid for an accurate reading. It **must not** sag.

Attach an indicator to the crankshaft as shown.

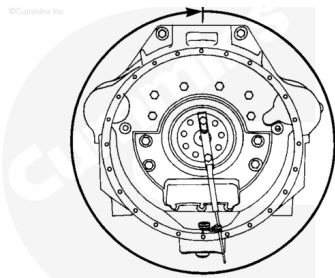


fh4hsta

Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero. Rotate the crankshaft one complete revolution, 360 degrees.

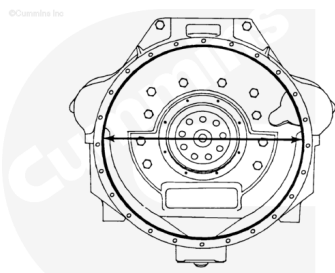
Record the total indicator runout.

The indicator **must** return to zero when returning to the start point.



fh4hstb

The maximum allowable total indicator runout depends on the diameter of the bore.



fh4hsga

Bore Diameter/Maximum Total Indicator Runout				
	mm		in	
No. 01	0.20 mm [0.008 in]	511	MIN	20.12
	511.2	MAX	20.13	

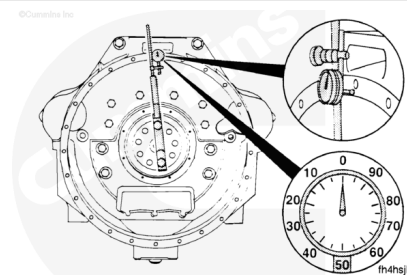
No. 0	0.25 mm [0.010 in]	647.7	MIN	25.50
	648.0	MAX	25.51	

If the alignment is **not** within specifications and the bore is round, the housing can be shifted.

If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.

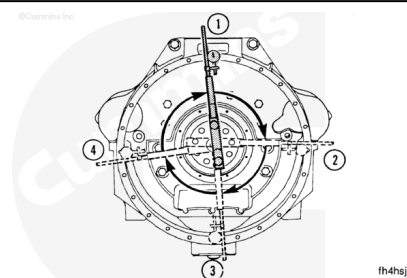
The crankshaft end clearance **must** be pushed or pulled in the same direction each time a point is measured.

Attach an indicator as shown. Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero.



Record the indicator reading at three different points; for example, 3 o'clock, 6 o'clock, and 9 o'clock.

Turn backward to the original position. Make sure the needle still points to zero. Determine the total indicator runout.

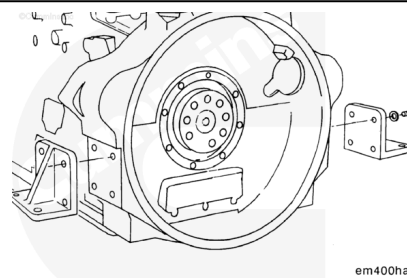


		mm	in
Example:			
	3 o'clock (2)	0.00	[0.00]
	6 o'clock (3)	+0.08	[+0.003]
	9 o'clock (4)	-0.05	[-0.002]
Equal total indicator runout:		0.13	[0.005]

Install the rear engine support brackets, washers, and bolts.

Tighten the bolts.

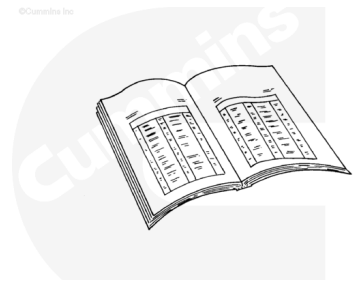
Torque Value: 360 n·m [265 ft-lb]



Finishing Steps

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ck800wa

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- Install the stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)
- Fill the lubricating oil and install the oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-020-tr.html>)
- Install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-024.html>)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/89/89-016-005.html>)
- Install the transmission, clutch, and all related components. See equipment manufacturer service

information.

- Connect the batteries or air supply to the air starter. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for proper operation.

Last Modified: 05-May-2015
